# A Generalization of a Definite Integral Involving an Exponential, a Polynomial, and an Inverse Trigonometric Function

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**Abstract**

In this note, we generalize a specific definite integral featured in the image file `dailyintegral-integral-medium-1x1-2026-08-09.jpg`. The original integral, which combines a quadratic polynomial, an exponential function, and the inverse sine function, is extended to a broader class of integrals involving arbitrary polynomials and scaled exponentials. We provide a reduction formula that maps the original transcendental integral to an algebraic one.

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## 1 Introduction

The problem sourced from `dailyintegral-integral-medium-1x1-2026-08-09.jpg` presents the following definite integral:

$$I = \int_0^1 x(x+3)e^x \sin^{-1}(x)\, dx \tag{1}$$

While this integral can be evaluated by applying integration by parts, its structure suggests a natural generalization to a wider family of integrals. We can replace the specific quadratic polynomial $P(x) = x(x+3)$ with an arbitrary polynomial $P(x)$ of degree $n$, and the exponential term $e^x$ with $e^{\lambda x}$.

## 2 The Generalized Integral Family

Let $P(x)$ be a polynomial of degree $n$, and let $\lambda \neq 0$ be a real constant. We define the generalized integral family as:

$$\mathcal{I}(P, \lambda) = \int_0^1 P(x)e^{\lambda x} \sin^{-1}(x)\, dx \tag{2}$$

## 3 Reduction Method

We proceed by applying integration by parts. Let:

$$u = \sin^{-1}(x) \implies du = \frac{1}{\sqrt{1-x^2}}\,dx \tag{3}$$

$$dv = P(x)e^{\lambda x}\,dx \tag{4}$$

To find $v$, we perform repeated integration by parts on $dv$ (often referred to as tabular integration). It is a standard result that the antiderivative of a polynomial multiplied by an exponential is given by:

$$v(x) = e^{\lambda x}\sum_{k=0}^{n}(-1)^k\frac{P^{(k)}(x)}{\lambda^{k+1}} \equiv Q(x) \tag{5}$$

where $P^{(k)}(x)$ denotes the $k$-th derivative of $P(x)$ with respect to $x$. Note that $Q(x)$ is simply $e^{\lambda x}$ multiplied by a new polynomial of degree $n$.

Applying the integration by parts formula $\int u\,dv = uv - \int v\,du$ to our generalized integral, we obtain:

$$\mathcal{I}(P,\lambda) = \left[Q(x)\sin^{-1}(x)\right]_0^1 - \int_0^1 \frac{Q(x)}{\sqrt{1-x^2}}\,dx \tag{6}$$

### Evaluating the Boundary Terms

Because of the bounds of integration, the evaluation of the $uv$ term is straightforward:

- At $x = 1$, $\sin^{-1}(1) = \frac{\pi}{2}$.
- At $x = 0$, $\sin^{-1}(0) = 0$.

Thus, the boundary evaluation simplifies to $\frac{\pi}{2}Q(1)$. The integral is fully reduced to:

$$\mathcal{I}(P,\lambda) = \frac{\pi}{2}Q(1) - \int_0^1 \frac{Q(x)}{\sqrt{1-x^2}}\,dx \tag{7}$$

This formula successfully reduces the transcendental-heavy integrand into a purely algebraic one (scaled by an exponential) over a square root.

## 4 Application to the Original Problem

We can easily apply this reduction to the specific integral from `dailyintegral-integral-medium-1x1-2026-08-09.jpg`.

Here, $\lambda = 1$ and $P(x) = x^2 + 3x$. The necessary derivatives of $P(x)$ are $P'(x) = 2x+3$ and $P''(x) = 2$.

The function $Q(x)$ becomes:

$$Q(x) = e^x\left[(x^2+3x) - (2x+3) + 2\right] = e^x(x^2+x-1) \tag{8}$$

Evaluating $Q(x)$ at $x = 1$ yields $Q(1) = e^1(1+1-1) = e$.

Applying our generalized reduction formula yields the immediate result:

$$I = \frac{\pi e}{2} - \int_0^1 \frac{e^x(x^2+x-1)}{\sqrt{1-x^2}}\,dx \tag{9}$$

From here, the resulting integral can be tackled utilizing series expansions or special functions depending on the level of precision required.

# 5 Further Examples of the Generalization

To demonstrate the broad applicability of the generalized reduction formula, we present two additional examples with varying polynomials and exponential scaling factors.

## Example 1: Constant Polynomial and Scaled Exponential

Consider the case where $P(x) = 1$ and $\lambda = 2$. We wish to evaluate:

$$\mathcal{I}(1, 2) = \int_0^1 e^{2x} \sin^{-1}(x)\, dx \tag{10}$$

Since $P(x) = 1$, its derivatives for $k \geq 1$ are identically zero. The function $Q(x)$ is simply:

$$Q(x) = e^{2x} \left(\frac{1}{2}\right) = \frac{e^{2x}}{2} \tag{11}$$

Evaluating $Q(x)$ at $x = 1$ yields $Q(1) = \frac{e^2}{2}$. Applying the reduction formula, we obtain:

$$\mathcal{I}(1, 2) = \frac{\pi e^2}{4} - \int_0^1 \frac{e^{2x}}{2\sqrt{1 - x^2}}\, dx \tag{12}$$

## Example 2: Linear Polynomial with a Negative Exponential

Next, consider $P(x) = x$ and $\lambda = -1$, yielding the integral:

$$\mathcal{I}(x, -1) = \int_0^1 x e^{-x} \sin^{-1}(x)\, dx \tag{13}$$

The derivatives of $P(x)$ are $P'(x) = 1$ and $P''(x) = 0$. Using the formula for $v(x)$, the function $Q(x)$ becomes:

$$Q(x) = e^{-x} \left[\frac{x}{-1} - \frac{1}{(-1)^2}\right] = -e^{-x}(x + 1) \tag{14}$$

Evaluating at the boundary $x = 1$ gives $Q(1) = -2e^{-1} = -\frac{2}{e}$. The reduced integral is then:

$$\mathcal{I}(x, -1) = -\frac{\pi}{e} + \int_0^1 \frac{e^{-x}(x + 1)}{\sqrt{1 - x^2}}\, dx \tag{15}$$

This clearly maps the original transcendental integral to a more manageable algebraic form over the root expression.